Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for: **Anaphylaxis in Children**

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SUMMARY

Management of anaphylaxis in children

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APPENDIX A - RESUSCITATION COUNCIL UK GUIDELINE 2021

Resuscitation Council UK

GUIDELINES 2021

Anaphylaxis

Anaphylaxis?

A = Airway B = Breathing C = Circulation D = Disability E = Exposure

Diagnosis – look for:

- Sudden onset of Airway and/or Breathing and/or Circulation problems[1]
- And usually skin changes (e.g. itchy rash)

Call for HELP
Call resuscitation team or ambulance

- Remove trigger if possible (e.g. stop any infusion)
- Lie patient flat (with or without legs elevated)
  - A sitting position may make breathing easier
  - If pregnant, lie on left side

Give intramuscular (IM) adrenaline[2]

Inject at **anterolateral aspect** – middle third of the thigh

- Establish airway
- Give high flow oxygen
- Apply monitoring: pulse oximetry, ECG, blood pressure

If no response:

- Repeat IM adrenaline after 5 minutes
- IV fluid bolus[3]

If no improvement in Breathing or Circulation problems[1] despite TWO doses of IM adrenaline:

- Confirm resuscitation team or ambulance has been called
- Follow REFRACTORY ANAPHYLAXIS ALGORITHM

1. Life-threatening problems

Airway
Hoarse voice, stridor

Breathing
↑work of breathing, wheeze, fatigue, cyanosis, SpO_2 <94%

Circulation
Low blood pressure, signs of shock, confusion, reduced consciousness

2. Intramuscular (IM) adrenaline
Use adrenaline at 1 mg/mL (1:1000) concentration

Adult and child >12 years:	500 micrograms IM (0.5 mL)
Child 6–12 years:	300 micrograms IM (0.3 mL)
Child 6 months to 6 years:	150 micrograms IM (0.15 mL)
Child <6 months:	100–150 micrograms IM (0.1–0.15 mL)

The above doses are for IM injection **only**.
Intravenous adrenaline for anaphylaxis to be given **only by experienced specialists** in an appropriate setting.

3. IV fluid challenge
Use crystalloid

Adults: 500–1000 mL
Children: 10 mL/kg

Resuscitation Council UK

GUIDELINES 2021

Refractory anaphylaxis

No improvement in respiratory or cardiovascular symptoms despite 2 appropriate doses of intramuscular adrenaline

Establish dedicated peripheral IV or IO access

Seek expert[1] help early
Critical care support is essential

Give rapid IV fluid bolus
e.g. 0.9% sodium chloride

&

Start adrenaline infusion
Adrenaline is essential for treating all aspects of anaphylaxis

Give IM* adrenaline every 5 minutes until adrenaline infusion has been started
*IV boluses of adrenaline are not recommended, but may be appropriate in some specialist settings (e.g. peri-operative) while an infusion is set up

Give high flow oxygen
Titrate to SpO_2 94–98%

Monitor HR, BP, pulse oximetry and ECG for cardiac arrhythmia
Take blood sample for mast cell tryptase

Follow local protocol
OR
Peripheral low-dose IV adrenaline infusion:
- 1 mg (1 mL of 1 mg/mL [1:1000]) adrenaline in 100 mL of 0.9% sodium chloride
- Prime and connect with an infusion pump via a dedicated line

DO NOT 'piggy back' on to another infusion line
DO NOT infuse on the same side as a BP cuff as this will interfere with the infusion and risk extravasation

- In both adults and children, start at 0.5–1.0 mL/kg/hour, and **titrate according to clinical response**
- Continuous monitoring and observation is mandatory
- ↑↑ BP is likely to indicate adrenaline overdose

Continue adrenaline infusion and treat ABC symptoms
Titrate according to clinical response

[1]Intravenous adrenaline for anaphylaxis to be given only by experienced specialists in an appropriate setting.

A = Airway
Partial upper airway obstruction/stridor:
Nebulised adrenaline (5mL of 1mg/mL)
Total upper airway obstruction:
Expert help needed, follow difficult airway algorithm

B = Breathing
Oxygenation is more important than intubation
If apnoeic:
- Bag mask ventilation
- Consider tracheal intubation

Severe/persistent bronchospasm:
- Nebulised salbutamol and ipratropium with oxygen
- Consider IV bolus and/or infusion of salbutamol or aminophylline
- Inhalational anaesthesia

C = Circulation
Give further fluid boluses and titrate to response:
Child 10 mL/kg per bolus
Adult 500–1000 mL per bolus
- Use glucose-free crystalloid (e.g. Hartmann's Solution, Plasma-Lyte®)

Large volumes may be required (e.g. 3–5 L in adults)
Place arterial cannula for continuous BP monitoring
Establish central venous access

IF REFRACTORY TO ADRENALINE INFUSION
Consider adding a second vasopressor **in addition** to adrenaline infusion:
- Noradrenaline, vasopressin or metaraminol
- In patients on beta-blockers, consider glucagon

Consider extracorporeal life support

Cardiac arrest – follow ALS ALGORITHM
- Start chest compressions early
- Use IV or IO adrenaline bolus (cardiac arrest protocol)
- Aggressive fluid resuscitation
- Consider prolonged resuscitation/extracorporeal CPR

APPENDIX B – DRUG DOSING

Adrenaline IM	1month-5yrs: 150micrograms (0.15ml of 1:1000) (1mg/1ml)adrenaline, repeated after 5 mins if required
	6-11 years: 300micrograms (0.3ml of 1:1000)(1mg/1ml), repeated after 5 mins if required
	>12 years: 500micrograms (0.5ml of 1:1000) (1mg/1ml), repeated after 5 mins if required
Adrenaline Nebuliser	400micrograms/kg of 1:1000(1mg/ml) adrenaline (max 5mg) NEB
Cetirizine PO	1 year: 250micrograms/kg twice daily
	2-5 years: 2.5mg twice daily
	6-11 years: 5mg twice daily
	>12 years: 10mg once daily
Loratadine PO	2-11 years (<31kg): 5mg once daily
	2 years -17 years (≥31kg): 10mg once daily
Prednisolone PO	All ages: 2mg/kg (max 40mg) once daily
Salbutamol Nebuliser	<5 years: 2.5mg NEB half hourly continue until reviewed
	>5 years: 5mg NEB half hourly continue until reviewed

For discharge:

Weight	**Adrenaline auto injector dose(micrograms)** **Prescribe by brand**
Child up to 15kg	Epipen Junior 150 micrograms / Jext 150 micrograms
Child 15-24kg	Epipen Junior 150 micrograms /Jext 150 micrograms. On the basis of 10mcg/kg/dose Epipen 300 micrograms / Jext 300 micrograms may be more appropriate for some children
≥25kg	Epipen 300 micrograms / Jext 300 micrograms

1. INTRODUCTION

1.1. Anaphylaxis is a severe, life threatening, generalised or systemic hypersensitivity reaction which is likely when both of the following criteria are met:

- Sudden onset and rapid progression of symptoms
- Life threatening airway and/or breathing and/or circulation problems

1.2. Anaphylaxis may be immunologically mediated, non-immunologically mediated or idiopathic. A significant proportion of anaphylaxis is classified as idiopathic in which no readily identifiable cause is found. Food is the commonest trigger for anaphylaxis in children. Insect venoms, drugs and latex are the other causes.

- Peanuts, tree nuts, fish, shellfish, cow's milk, soya and egg are the most common food in anaphylaxis.
- In IgE mediated allergic anaphylaxis, there is systemic release of mediators from mast cells and basophils.

1.3. Allergy is an immune-mediated hypersensitivity reaction to a particular allergen resulting in a pathological reaction. These are usually common environmental substances, food or drugs. Allergic reactions can be local or systemic, and may be mild, moderate or severe.

2. IMPLEMENTATION

2.1 This guideline will be implemented in all paediatric areas such as the paediatric emergency department, inpatient ward, outpatient ward and day care.

3. ROLES AND RESPONSIBILITIES

3.1 All medical and nursing staff in paediatrics responsible for managing children with anaphylaxis and allergies should ensure they are familiar with this guideline.

4. GUIDELINE – REFER TO APPENDIX A AND B (TREATMENT FLOW CHART AND MEDICATION DOSING) ABOVE

- Call for help. Assess airway, Breathing and Circulation. Remove the allergen if possible (e.g. bee sting).
- **The main treatment in anaphylaxis is IM adrenaline and this should be administered to all children as soon as anaphylaxis is suspected.**
- Make the patient lie in a supine position and elevate the legs. If there is any difficulty in breathing, they should be nursed in sitting position.
- **Adrenaline:** Adrenaline may be dosed according to weight or age of the patient. It is always given via the intramuscular route (IM).
  - Use solution containing adrenaline 1mg/ml (1:1000) or an adrenaline auto-injector where appropriate.
- If there is no response after 5 minutes, then the dose of IM adrenaline should be repeated.
- If the child has a cardiac or respiratory arrest, follow the arrest protocol as per Resus guidelines: https://www.resus.org.uk/library/2021-resuscitation-guidelines
- Children with suspected anaphylaxis should also:
  - Have cardiac monitoring attached & commenced on high flow O2 via non re-breather mask
  - Have IV access secured
  - Be observed closely for worsening of signs and symptoms
- **Senior medical help should be sought if there are any concerns.**

4.1 Refractory Anaphylaxis

Peripheral IV adrenaline infusion for refractory anaphylaxis (only to be given by experienced specialist in appropriate setting)

Preparation

• **Continuous monitoring and observation are mandatory**:

- ECG, pulse oximetry, non-invasive BP at least every 5 minutes

• Mix 1 mg (1 mL of 1 mg/ml [1:1 000]) adrenaline in **100 mL** of 0.9% sodium chloride and connect using an infusion pump via a dedicated line.

- **Do not "piggy-back" on to another line** unless using an anti-reflux valve.
- **Do not** infuse on the same side as a BP cuff, as BP measurements will interfere with the infusion and risk extravasation injury.

Initiation and adjustment

• **In children and adults, start at 0.5 – 1.0 mL/kg/h** depending on severity:

- Moderate severity 0.5 mL/kg/h (~0.1 micrograms/kg/min)
- Severe (hypotensive or hypoxic) 1 mL/kg/h

• **Titrate** up or down according to response, aiming for the lowest effective rate.

- Steady state is reached 5 – 10 min after a change in infusion rate.
- Monitor infusion site regularly to ensure patency of cannula

• **Tachycardia, tremor, pallor with a normal or raised BP** may indicate excessive adrenaline treatment: reduce the infusion rate (or stop infusion if severe).

• If refractory to adrenaline infusion, seek urgent further expert help. Patients will require central venous access for prolonged infusions; follow local protocols.

WEANING

• **As symptoms improve, reduce the infusion**, aiming for 50% of the starting rate.

• One hour after resolution of all symptoms and signs, reduce the infusion rate progressively

5. AIRWAY/ STRIDOR

- Children with anaphylaxis can develop airway compromise from laryngeal oedema
- This should be suspected if the child has stridor with other allergic signs
- Nebulised Adrenaline 0.4 mg/kg of 1:1000 (1mg/1ml) up to a max of (5mg = 5mls) made up to total volume of 5 mls with 0.9% sodium chloride should be given. There is usually a rapid response to nebulised adrenaline. This is in addition to the IM adrenaline they should have also received. The child should have continuous cardiac monitoring.

- Children can develop rebound worsening of their oedema once the effect of the adrenaline nebuliser has worn off. This should be anticipated and a second adrenaline nebuliser should be given in such cases. Anaesthetic & ITU staff should be called urgently to secure the airway.

6. BREATHING/WHEEZE

- Children with anaphylaxis can have a bronchospasm. This is more likely in children with pre-existing asthma but can occur in any child.
- Bronchospasm should be suspected if clinically the child has a wheeze, respiratory distress or reduced oxygen saturations.
- Treatment is with nebulised Salbutamol (2.5 mg if < 5 years, 5 mg if > 5 years) **in addition to IM Adrenaline**. If the wheeze persists, repeat the salbutamol nebuliser and follow the Acute Asthma guideline.
- **Corticosteroids may be indicated where an acute asthma exacerbation may have contributed to the severity of anaphylaxis. Steroids should be given by the oral route where possible.**
- If the child fails to improve with these measures, then ITU & anaesthetic teams should be called to review the child, and the child should be treated as per the **Acute Asthma guideline**. https://www.brit-thoracic.org.uk/standards-of-care/guidelines/btssign-british-guideline-on-the-management-of-asthma/

7. CIRCULATION/HYPOTENSION/COLLAPSE

- Children with anaphylaxis can develop severe hypotension and shock following generalised capillary leakage.
- Signs of shock should be actively sought in any child presenting with anaphylaxis with regular observations including BP measurements, in addition to clinical assessment.
- Any child with features of shock (tachycardia, hypotension, prolonged capillary refill) should have IV access obtained. If IV access is not easily obtained, then the Intra osseous route should be considered promptly.
- Once IV access is secured, an initial fluid bolus of 10 ml/kg sodium chloride 0.9% should be given.
- Following reassessment, if additional fluid boluses are required then the use of Plasma-Lyte® solution **may** be considered instead of sodium chloride 0.9%.
- If a total of >40 ml/kg fluid is required to correct shock, then ITU and PICU advice should be sought as the child may require elective intubation & positive pressure ventilation to reduce the risk of pulmonary oedema. If the child remains shocked however, they should still be treated with fluid boluses as clinically indicated even if this exceeds 40ml/kg.

- If there is on-going hypotension & shock, then the use of an IV Adrenaline infusion should be considered and PICU contacted for advice & further management.

8. INVESTIGATIONS

8.1 Mast cell tryptase

Consider sending blood samples for mast cell tryptase if the cause is thought to be venom-related, drug-related or idiopathic. (Measuring tryptase levels should not delay initial resuscitation).

- 1st sample (clotted bottle , at least 2mls of blood) taken as soon as possible after emergency treatment has started
- 2nd sample ideally within 1–2 hours (but no later than 4 hours) from the onset of symptoms
- Inform the parents that a blood sample may be required at follow-up in the allergy clinic to measure baseline mast cell tryptase levels
- Tryptase concentration in the blood may not increase significantly until 30 minutes or more after the onset of symptoms, and peak 1-2 hours after onset. The half-life of tryptase is short (approximately 2 hours), and the concentration may be back to normal within 6-8 hours, so timing of any blood samples is very important.

9. DISCHARGE

9.1 **Discharge criteria laid out by the resuscitation council can be found below. All patients should be warned of the signs of biphasic reactions.** Do not discharge if there are respiratory or cardiovascular symptoms or signs.

Consider fast-track discharge (after 2 hours observation from resolution of anaphylaxis) if:	Minimum 6 hours observation after resolution of symptoms recommended if:	Observation for at least 12 hours following resolution of symptoms if any one of the following:
• Good response (within 5–10 minutes) to a single dose of adrenaline given within 30 minutes of onset of reaction **and** • Complete resolution of symptoms **and** • The patient already has unused adrenaline auto-injectors and has been trained how to use them **and** • There is adequate supervision following discharge	• 2 doses of IM adrenaline needed to treat reaction* **or** • Previous biphasic reaction	• Severe reaction requiring >2 doses of adrenaline. • Patient has severe asthma or reaction involved severe respiratory compromise. • Possibility of continuing absorption of allergen, e.g. slow-release medicines. • Patient presents late at night, or may not be able to respond to any deterioration. • Patients in areas where access to emergency care is difficult.
In all cases, discharge must comply with NICE Clinical Guidance CG134.[96]		

*It may be reasonable for some patients to be discharged after 2 hours despite needing two doses of IM adrenaline, e.g. following a supervised allergy challenge in a specialist setting.

BSACI Action plan available for children with mild to moderate allergies that do not need auto-injectors: http://www.bsaci.org/about/download-paediatric-allergy-action-plans.

9.2 **Admission** should be considered if any of the following circumstances apply:

- Greater than one dose of adrenaline (including nebulised adrenaline) required.
- A fluid bolus required.
- Inadequate response to treatment.
- The child lives a long distance from medical services.
- Severe anaphylactic reaction
- Under 1 year of age
- Moderately severe reaction with trigger unknown
- Failure to thrive associated with atopy
- Severe skin or gut symptoms associated with allergic reaction

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APPENDIX C – DISCHARGE CRITERIA

Patients to be discharged after appropriate observation once all of the following criteria have been met:

Discharge criteria –To be signed off by medical or nursing staff	**Date criteria met**
Identification of potential allergen A reasonable assessment of the potential allergen has been made and the family are advised to avoid until further assessment by an allergist	
Assessment of clinical status Respiratory and cardiovascular symptoms have completely resolved	
Advice to parents The parents have been advised on the signs of anaphylaxis and when to call an ambulance, they should be advised of the possibility of biphasic reactions. They should have been given written information pertaining to this; anaphylaxis management plans are available from the **BSACI website.** http://www.bsaci.org/about/pag-allergy-action-plans-for-children	
Antihistamine prescription Parents are given a non sedating anti-histamines to take home.	
Antihistamine and auto injector prescription All patients presenting with an episode of anaphylaxis should be provided with two adrenaline auto-injectors on discharge and a copy of BSACI action plan.	
Adrenaline auto injector training **Parents should be trained in using the adrenaline auto injectors by a trained nurse or doctor. Training video available at www.itchysneezywheezy.co.uk**	
Not for discharge until reviewed by the allergy team if 1. Severe anaphylactic reaction – adrenaline infusion or 2 or more doses of IM adrenaline 2. Under 1 year of age 3. Moderately severe reaction with trigger unknown 4. Patient has concurrent Failure to Thrive 5. Severe skin or gut symptoms associated with atopy	
GP advice The discharge letter informing the GP that a referral to the paediatric allergy service has been done for further assessment.	